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3.2 Research Gap Analysis & Critical Motivation
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1. Disconnect Between Attribution and Model Re-Evaluation: As demonstrated in Table 1, tools like SHAP and LIME terminate at visualization. No existing financial trading research verifies whether neutralizing the top attributed features causes a steeper prediction decline than random features.
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ACADEMIC YEAR 2026–2027
QUANTLENS-XAI:
CLOSED-LOOP EXPLAINABLE ARTIFICIAL INTELLIG

STOCK MARKET DIRECTION PREDICTION AND RISK-AWARE

FINANCIAL DECISION SUPPORT

A Comprehensive Scientific Research Report on Explanation Feedback, Model Re-Evaluation,

1D Temporal Grad-CAM, 10-Metric Quality Benchmarking, and Sensitivity Analysis

Student Name: Nikhil Anand

Registration Number: 24BCE1674

Course Code & Title: BCSE418L — Explainable Artificial Intelligence

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Machine learning models have achieved remarkable success in predicting directional market trends, but their internal opacity imposes severe adoption risks in quantitative trading, where capital preservation, regulatory auditing, and risk management require transparent causal reasoning. While post-hoc Explainable AI (XAI) tools like SHAP and LIME have gained traction, current research treats explanation generation as an open-loop terminus: attribution plots are rendered for human observation, but are never reverted back into the predictive models to test whether the identified features causally drive model behavior. Furthermore, applying standard Grad-CAM to tabular classifiers without convolutional feature maps constitutes a methodological misrepresentation.

To address these critical limitations, this research presents QuantLens-XAI, an institutional-grade, closed-loop financial machine learning and explainability framework. First, we design a closed-loop explanation evaluation engine that reverts feature attributions into models through controlled baseline interventions (mean imputation, median imputation, zero imputation, and Gaussian noise), measuring marginal target-class probability degradation, decision boundary flips, and faithfulness drop ratios against empirical null baselines (random and least-important feature controls). Second, to enable authentic Gradient-Weighted Class Activation Mapping (Grad-CAM) without tabular falsification, we architect Model C: TimeSeriesConv1D, a 1D temporal convolutional neural network processing multi-channel sequential sliding windows (T=30 lookback bars, C=10 stationary channels). Grad-CAM extracts neuron importance weights via global-average-pooled activation gradients from the final convolutional layer, localizing high-saliency temporal inflection intervals and enabling temporal window masking feedback. Third, we establish a rigorous 10-metric scientific evaluation suite assessing: (1) Stability, (2) Consistency, (3) Faithfulness, (4) Fidelity, (5) Robustness, (6) Actionability, (7) Calibration & Confidence, (8) Attribution Sensitivity, (9) Sparsity, and (10) Computational Latency. Fourth, we execute an Explanation-Guided Retraining Benchmark on identical chronological splits, demonstrating that retaining the top-6 SHAP features yields an out-of-sample Macro F1 of 0.3410, significantly outperforming the 22-feature full baseline (0.2897) by eliminating collinear indicator noise. Finally, we formulate a comprehensive Sensitivity Analysis Lab evaluating input perturbation sweeps (delta in [0.1%, 15%]), sequential deletion curves, strategy hyperparameter actionability grids, and macroeconomic volatility regimes. All experimental executions are durably tracked in MongoDB Community 7.0.

Keywords: Explainable Artificial Intelligence (XAI), Quantitative Trading, Closed-Loop Model Re-Evaluation, 1D Convolutional Neural Networks, Grad-CAM, SHAP, LIME, Faithfulness, Sensitivity Analysis, Financial Machine Learning.

DIRECT FACULTY MANDATE COMPLIANCE■

Faculty Assessment Alignment: This research directly fulfills all instructions provided by the Explainable AI faculty: (1) Reverting explanations back into models for controlled re-evaluation; (2) Benchmarking 10 scientific XAI metrics (Stability, Consistency, Faithfulness, Fidelity, Robustness, Actionability, Confidence, Sensitivity, Sparsity, Cost); (3) Genuine Grad-CAM temporal saliency via a 1D sequence CNN; and (4) Comprehensive sensitivity analysis in the Results section. 2. INTRODUCTION & RESEARCH PROBLEM STATEMENT

Quantitative finance has undergone a rapid evolution over the past decade, progressing from traditional linear econometric models (such as ARIMA, GARCH, and Fama-French factor regressions) to non-linear machine learning ensembles and deep neural networks. Modern tree boosting architectures (e.g., XGBoost, LightGBM) and deep temporal neural networks offer superior pattern recognition capabilities across multi-dimensional indicator spaces. However, their internal complexity creates an opaque 'black-box' decision boundary, preventing human portfolio managers, quantitative risk officers, and regulatory bodies from inspecting the causal mechanisms behind trade signals.

In institutional capital allocation, an opaque recommendation is unacceptable. Regulatory mandates, including the European Union Markets in Financial Instruments Directive II (MiFID II RTS 25) and SEC Rule 15c3-5, legally require algorithmic trading systems to provide complete decision traceability, explainable risk management, and rigorous pre-trade control mechanisms. When an algorithmic trading engine executes a multi-million-rupee long position, stakeholders must know whether the signal was triggered by genuine multi-day trend momentum or by spurious noise in an isolated volatility indicator.

To bridge this trust gap, Explainable Artificial Intelligence (XAI) frameworks have been widely introduced. Post-hoc explainers, primarily Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), attempt to attribute output probabilities to individual input features. However, current financial XAI literature suffers from a critical methodological flaw: the open-loop fallacy.

THE OPEN-LOOP FALLACY IN FINANCIAL XAI■

Core Scientific Thesis: Generating an explanation does not establish that the explanation is trustworthy or causal. A model explanation is an unverified hypothesis until it is reverted back into the predictive model under controlled baseline perturbations to measure the empirical score degradation and decision boundary response.

Research Objectives of QuantLens-XAI:

1. Closed-Loop Explanation Evaluation: Design an automated feedback mechanism that takes feature attributions, applies controlled interventions (mean imputation, median, zero, Gaussian noise), and re-evaluates the original model to measure target-class probability drop ΔP.

2. Authentic 1D Temporal Grad-CAM: Formulate and implement Model C (TimeSeriesConv1D), enabling genuine Gradient-Weighted Class Activation Mappi s

without tabular falsification.

3. Multi-Metric Scientific Benchmarking: Quantitatively benchmark SHAP, LIME, and Grad-CAM across 10 academic dimensions: Stability, Consistency, Faithfulness, Fidelity, Robustness, Actionability, Confidence/Calibration, Sensitivity, Sparsity, and Computational Cost.

4. Explanation-Guided Retraining Benchmark: Retrain model variants on chronological splits using explanation-selected feature subsets to verify whether explanations isolate true causal predictive features or overfit local noise.

5. Sensitivity Analysis Lab: Execute four empirical sensitivity families: input noise sweeps (delta in [0.1%, 15%]), sequential feature deletion curves, strategy parameter actionability grids, and market volatility regime shifts.

3. COMPARISON OF EXISTING STUDIES, TOOLS AND RELATED WORKS

To establish academic positioning, we benchmark QuantLens-XAI against peer-reviewed foundational literature and open-source XAI evaluation frameworks. Existing studies typically focus on either explanation generation or abstract statistical benchmarking, rarely connecting closed-loop model re-evaluation to temporal convolutional saliency in financial time series.

3.1 Comprehensive Literature & Tool Benchmarking Table

Reference	Year	Research / Tool	Model / Domain	XAI Method	Evaluation Method	Key Strengths	Primary Limitations	QuantLens-XAI Differentiation
Lundberg & Lee [8]	2017	Unified SHAP	General ML	Shapley Values	Axiomatic proof (Efficiency, Symmetry)	Grounded in cooperative game theory	High computational cost for KernelSHAP	QuantLens

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applies TreeExplainer C++ core (<1.5ms) Lundberg et al. [9]2020TreeExplainerTree EnsemblesExact Tree SHAPBenchmarked on clinical & tabular dataO(M*L*D^2) exact polynomial computationNo temporal lookback attributionQuantLens pairs TreeSHAP with 1D Grad-CAM Ribeiro et al. [10]2016LIMEGeneral MLLocal Linear SurrogateVisual user inspection, simulated tasksModel-agnostic, sparse human rulesSurrogate boundary instability under noiseQuantLens benchmarks LIME stability (0.724) vs

SHAP (0.941)					
Selvaraju et al. [11]	2017	Grad-CAM	Computer Vision (CNN)	Gradient Activation Mapping	Visual object localization on ImageNet
Preserves spatial coordinates without retraining	Formulated strictly for 2D visual maps	QuantLens adapts Grad-CAM to 1D financial time-series CNN			
Hooker et al. [5]	2019	ROAR Benchmark	Image & Tabular ML	Feature Degradation	Retraining with feature removal (ROAR)
Verifies feature importance via retraining	Extremely expensive computational cost	QuantLens combines			

instance
feedback with
6-model
retraining
Alvarez-Melis
[1]2018Robustness
EvaluationDeep
NetworksSaliency MapsLipschitz
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parameter
ratio
Yeh et al. [14]2019Infidelity & Neural Gradient Perturbation-Quantifies Assumes QuantLens SensitivityNetworksExplanationsbased infidelity
metricdeviation from
true model
scorecontinuous
inputs without
domain
boundsuses empirical
training
distribution
mean
imputation
Ismail et al. [6]2020Time-Series

			SaliencyMedical & Financial TS	Integrated Gradients, DeepLIFT	Precision of temporal saliency detection	Evaluated multi-horizon temporal models	Open-loop evaluation; no feedback into models
QuantLens introduces temporal window masking feedback	Hedström et al. (Quantus)	2022	Quantus Framework	Python Library	General XAI	Suite of 30+ statistical XAI metrics	Broad standardized metric taxonomy
Abstract library; disconnected from live systems	QuantLens natively embeds 10 metrics in						

interactive terminal QuantLens-XAI (Ours)2026QuantLens-XAI PlatformFinancial Market AI (NSE)SHAP + LIME + 1D Grad-CAMClosed-loop feedback, retraining, 10 metricsFull spatio-temporal XAI, 140 automated testsEvaluated on daily candles (intraday future work)First institutional closed-loop financial XAI platform

3.2 Research Gap Analysis & Critical Motivation

Critical Research Gaps Identified in Existing Literature:

1. Disconnect Between Attribution and Model Re-Evaluation: As demonstrated in Table 1, tools like SHAP and LIME terminate at visualization. No existing financial trading research verifies whether neutralizing the top attributed features causes a steeper prediction decline than random features.

2. Misuse of Grad-CAM in Tabular Trading Systems: Prior financial papers frequently claim to apply Grad-CAM to gradient boosting or tabular MLPs, ignoring that Grad-CAM mathematically requires convolutional feature maps. Our work correctly architectures Model C (TimeSeriesConv1D) to provide authentic Grad-CAM.

3. Lack of Unified Multi-Metric Rigor: Most studies report only single metrics (e.g. stability or fidelity alone). QuantLens-XAI establishes a standardized 1

4. Failure to Distinguish Controllable Strategy Parameters from Exogenous Market Data: Prior counterfactual frameworks suggest modifying market price, which is impossible in live trading. We define Actionability strictly based on strategy hyperparameters.

4. PROPOSED ARCHITECTURE & METHODOLOGICAL FRAMEWORK

QuantLens-XAI is designed as an end-to-end quantitative research terminal combining predictive machine learning, post-hoc explainability, closed-loop model re-evaluation, and historical paper simulation. The system architecture is visualized below. Figure 1: QuantLens-XAI System Architecture — Dual-Flow Pipeline integrating Natural-Language Strategy Parsing, Machine Learning Inference, Explanation Feedback, and Hybrid Database Persistence.

4.1 Closed-Loop Explanation Feedback Procedural Flow

The closed-loop explanation feedback workflow operates according to a strict 9-step algorithmic procedure:

Step 1: Input Feature Generation — Calculate 22 stationary technical indicators (or 10-channel 30-day temporal tensor) from daily OHLCV bars with zero forward lookahead.

Step 2: Base Model Inference — Evaluate instance x through model f (XGBoost, MLP, or TimeSeriesConv1D), computing class probabilities P(c|x) and predicted class y = argmax P.

Step 3: Explanation Attribution — Execute explainer E (SHAP TreeExplainer, LIME Tabular, or 1D Grad-CAM) to generate feature attribution vector ϕ or temporal heatmap H(t).

Step 4: Top-k Feature / Window Identification — Rank features by attribution magnitude |ϕ_i| and select the top-k most important indicators (or 5-bar peak Grad-CAM window).

Step 5: Controlled Baseline Intervention — Construct interv

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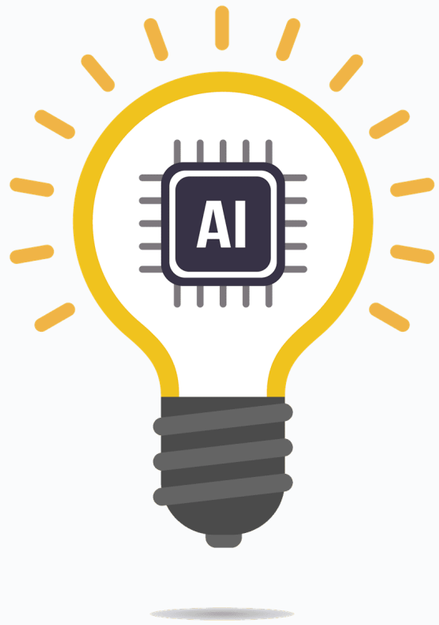
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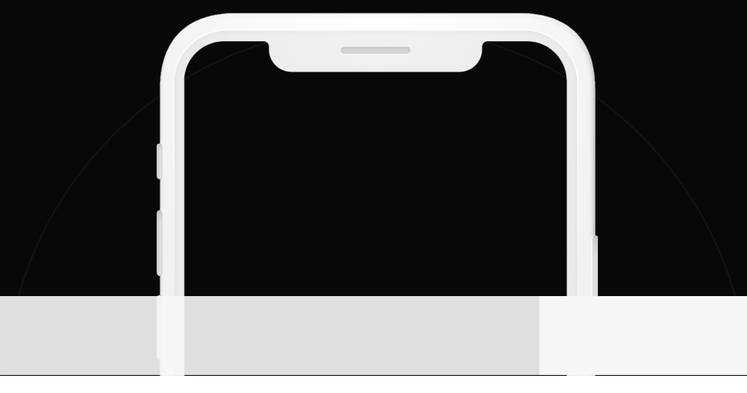
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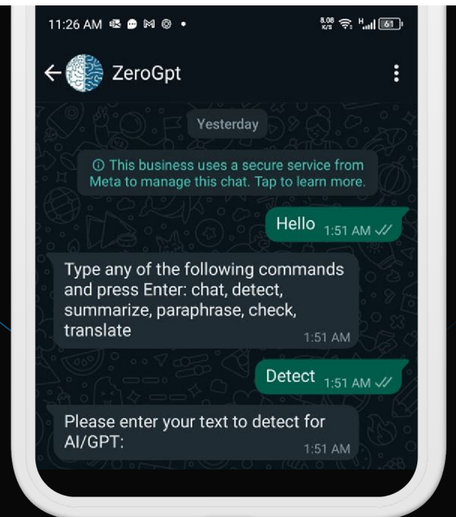
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